

PERSPECTIVE

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Rethinking ethical academic integrity ecosystems in higher education in the age of artificial intelligence

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Abstract

This Perspective paper argues that generative artificial intelligence (AI) is transforming the foundations of academic integrity in higher education, not by encouraging individual misconduct but by exposing systemic weaknesses in policy design, assessment practices, and institutional culture. Drawing on sociocultural and cognitive theories, the paper introduces the Higher Education Integrity Ecosystem Model, a conceptual framework that explains integrity as an emergent property of four interacting layers: governance and policy, assessment design, institutional and cultural norms, and individual moral agency. Unlike existing models that emphasize individual responsibility or organizational compliance, this perspective positions AI as a transformative force that reshapes authorship, authenticity, and ethical decision-making across all layers of the system. The paper argues that sustaining integrity in the Age of AI requires systemic redesign rather than incremental adjustments. Key implications for practice include developing transparent, process-oriented AI-use policies; redesigning assessments to emphasize human judgment and learning processes; cultivating institutional cultures that normalize disclosure and responsible AI use; and supporting students in exercising moral agency when interacting with increasingly capable AI tools. By aligning integrity governance with the realities of ubiquitous AI, the model offers a pathway toward ethical sustainability that reinforces fairness, trust, and human agency. The paper also highlights connections to the United Nations Sustainable Development Goals, particularly Quality Education (SDG 4) and strong, just institutions (SDG 16).

Keywords Academic integrity, AI, Higher education assessment, AI ethics, Educational governance, Assessment equity

1 Introduction

Integrity in higher education (HE) is not only an ethical concern but also a sustainability issue that shapes the long-term trust, equity, and resilience of educational systems. As universities increasingly adopt online learning and artificial intelligence (AI) technologies, questions about fairness, accountability, and moral responsibility have become central to the sustainability of learning environments. Recent reviews reveal a notable rise



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in academic misconduct between 2013 and 2023 [36], reflecting how institutional structures and assessment practices often fail to support ethical engagement. Practices such as using ChatGPT to generate essays or exchanging answers during online examinations are frequently rationalized as adaptive responses to poor assessment design [31]. Yet such justifications blur the distinction between legitimate learning strategies and academic dishonesty, eroding the moral foundations of higher education governance. What distinguishes the current moment is that generative AI is not merely another technological tool but a paradigm-shifting force that collapses traditional boundaries between human and machine authorship, alters the nature of cognitive labor, and challenges long-standing assumptions about originality, authorship, and assessment integrity.

Academic dishonesty—defined as the deliberate pursuit of unfair advantage [9]—has expanded in form and complexity. It now includes AI-assisted content generation, unapproved collaboration, and subtle forms of plagiarism, often triggered by performance pressure, inequitable access to resources, and opaque institutional policies [28]. These behaviors reveal systemic shortcomings rather than individual moral failure: assessment designs that privilege memorization over application, and governance systems that emphasize accountability over learning, generate environments in which misconduct appears rational. High-stakes testing and inconsistent honor codes further entrench inequity, prioritizing ranking over growth [15].

This Perspective argues that such integrity challenges are best understood through a sustainability lens—one that views ethical behavior as the product of coherent systems rather than isolated acts of compliance. Sustainable education extends beyond environmental dimensions; it encompasses ethical, cultural, and institutional sustainability. Ethical sustainability requires that learners and educators operate within systems that nurture fairness, transparency, and trust over time. Drawing on sociocultural [40] and cognitive theories [29], this paper advances the *Higher Education Integrity Ecosystem Model*, a framework that reimagines assessment as a governance-embedded, ethically sustainable process.

Language assessment provides an illustrative case, as cultural bias and linguistic inequities often exacerbate opportunities for misconduct [35]. Yet the same dynamics pervade other disciplines, where AI and digital tools further complicate what counts as “authentic” work. Addressing these challenges requires a shift from punitive enforcement to ethical system design—where governance, pedagogy, and learner agency interact to promote moral responsibility. Within the context of sustainability, integrity is conceptualized not as a static rule-based principle but as a living practice embedded in institutional cultures that foster equitable participation and long-term ethical development.

By linking academic ethics to sustainability, this paper positions integrity as essential to the social contract of higher education. It proposes that cultivating sustainable integrity ecosystems—grounded in transparency, collaboration, and trust—is key to maintaining both academic credibility and societal trust in higher education credentials.

Although previous ecological and systemic models of academic integrity have emphasized the interplay between institutional policy, assessment design, and learner behavior [1, 18], existing frameworks tend to focus either on individual responsibility or on organizational compliance. These approaches provide valuable insight into cultural and structural influences, yet they often lack a sustainability-oriented perspective that

accounts for the dynamic co-evolution of AI technologies, governance structures, and ethical culture. Current models also overlook the growing ambiguity introduced by generative AI—particularly the blurred boundaries between academic support and misconduct—and the ethical tensions related to algorithmic bias and digital inequities [10, 26]. This Perspective paper addresses these gaps by proposing the Higher Education Integrity Ecosystem Model, a framework that reconceptualizes integrity as an emergent property of sustainable system design. The model contributes a novel synthesis by integrating sociocultural and cognitive theories with principles of ethical sustainability, thereby extending earlier ecological approaches and offering a governance-oriented lens suited to AI-mediated higher education environments.

This paper advances a distinctive perspective by arguing that academic integrity in the age of AI is not merely an individual behavioral issue but a matter of systemic ethical sustainability. It contends that integrity emerges from the interaction of governance, assessment design, institutional culture, and learner agency, and that existing models insufficiently capture how these elements co-evolve alongside rapidly advancing AI technologies. By proposing the Higher Education Integrity Ecosystem Model, this Perspective reframes integrity as an adaptive, sustainability-driven system and offers a novel conceptual lens for understanding how universities can cultivate ethically resilient learning environments in AI-mediated higher education. The present paper adopts a conceptual and integrative methodological approach appropriate for a Perspective article. Rather than reporting original empirical findings, the analysis synthesizes existing scholarship across academic integrity research, educational governance, sociocultural theory, and the emerging literature on AI in higher education. The aim is to develop a theoretically informed framework—the Higher Education Integrity Ecosystem Model—by drawing connections between current debates, identifying gaps in existing systemic or ecological approaches, and offering a novel conceptual lens for understanding integrity in the Age of AI. No human participants were involved, and no primary data were collected.

2 Reframing cheating in higher education: an ethical sustainability challenge

Academic misconduct in higher education (HE) is often discussed as a moral lapse or an outcome of individual choice. Yet, from a sustainability perspective, cheating reflects deeper institutional and cultural fragilities that undermine the resilience of educational ecosystems. When systems prioritize performance, competition, and efficiency over equity and ethical growth, dishonesty becomes not an exception but a predictable byproduct of structural imbalance. Understanding cheating as a sustainability issue demands attention to how educational governance, pedagogy, and culture interact to either sustain or erode integrity over time.

Vygotsky's [40] sociocultural theory reminds us that learning occurs through social mediation and guided participation. Ethical scaffolding—teacher feedback, peer collaboration, formative assessment—is central to sustainable education because it nurtures learners' moral and intellectual autonomy. However, when such support is replaced or distorted by unauthorized assistance, such as AI-assisted plagiarism, the learning ecosystem's equilibrium is disrupted [27]. What was meant to extend learners' capacity becomes a tool for avoidance, revealing an erosion of trust and a breakdown in the ethical reciprocity between students and institutions.

From a cognitive standpoint, self-concept maintenance theory offers insight into the psychological dimensions of this imbalance. Individuals rationalize unethical behavior—“everyone cheats,” “the system is unfair”—to preserve a positive moral identity under pressure [29]. Yet this internal dissonance is magnified by institutional failures: vague assessment criteria, excessive workloads, and limited ethical dialogue reinforce perceptions of injustice [23]. When governance neglects to cultivate integrity through policy, students interpret compliance as survival rather than responsibility.

To strengthen cohesion, these sociocultural and cognitive perspectives can be viewed as complementary expressions of the same sustainability breakdown. In contexts where ethical scaffolding is weak, workloads are misaligned, and expectations are unclear, students increasingly use cheating—including AI-mediated shortcuts—as a coping response to structural pressures rather than an intentional rejection of ethical norms. The Higher Education Integrity Ecosystem Model proposed in this paper integrates these sociocultural and cognitive insights into a unified sustainability framework. It conceptualizes academic integrity as a continuum ranging from ethical reinforcement—such as transparent governance and integrity-based orientation programs—to explicit misconduct, including the covert use of AI tools. Between these poles lie ambiguous zones of behavior, such as unmonitored peer collaboration or overreliance on automated feedback, which highlight the structural vulnerabilities of contemporary higher education. These gray areas are particularly salient for international students navigating cultural incongruities and digital inequities [35].

A more cohesive view also positions structural inequities and digital divides not as parallel issues but as core mechanisms of sustainability failure. When access to technology, digital literacy, or AI fluency is uneven, students experience fundamentally different ethical terrains. Those with greater technological capital leverage AI to manage workloads or compensate for linguistic disadvantage, while others face heightened pressure, marginalization, or exclusion. Recent work further demonstrates that inequitable access to advanced AI systems—such as paid large language models, specialized writing tools, or institutionally licensed platforms—creates stratified learning environments where privileged students gain disproportionate cognitive and academic advantages [37]. As a result, integrity challenges are not evenly distributed but mapped onto existing social, economic, and linguistic inequalities. In such conditions, cheating becomes a symptom of unequal participation rather than purely an individual transgression.

Framing cheating as a coping mechanism [43] risks legitimizing unsustainable systems that perpetuate inequality. The emergence of generative AI intensifies these systemic vulnerabilities in unprecedented ways. Unlike previous technologies that supported specific tasks, AI now performs core academic competencies—summarizing, synthesizing, argument-building, language production, and data interpretation—at scale and with minimal friction. This shift complicates not only how cheating is defined but also how learning itself is conceptualized. In the Age of AI, students can bypass foundational cognitive processes without intentional misconduct, simply by interacting with tools that automate reasoning. This capability destabilizes traditional indicators of learning, amplifies uncertainty about authorship, and exposes integrity systems that were designed for a pre-AI educational landscape. Students with greater access to digital tools and technological literacy gain disproportionate advantages [25], while others remain marginalized. Taken together, these inequities illustrate how structural pressures, coping responses,

and technological divides converge into a system-level integrity risk rather than functioning as isolated factors. Recognizing these dynamics shifts the ethical conversation from individual accountability to institutional sustainability. A resilient academic ecosystem ensures that policies, assessments, and technologies align with principles of distributive justice, relational trust, and moral education.

Unlike conventional theoretical approaches that treat integrity as an isolated virtue, the ecosystem perspective views it as a shared ethical infrastructure. Integrity becomes a form of ethical resilience, sustained through ongoing alignment between structure, culture, and individual agency. By organizing the diverse drivers of misconduct around a central sustainability framework, this section establishes the conceptual foundation for the proposed model and clarifies why integrity must be addressed systemically rather than through isolated interventions. This reorientation not only deepens theoretical understanding but also opens practical pathways for designing assessments that sustain fairness, authenticity, and moral growth in the digital age.

3 Drivers of cheating: systemic and cultural pressures in higher education

Academic dishonesty does not emerge in a vacuum; it is a systemic response to institutional and cultural conditions that often contradict the values of ethical and sustainable education. From a sustainability standpoint, cheating signifies a loss of equilibrium between equity, transparency, and trust—the moral resources that sustain academic communities. When educational systems reward outcomes over process, and compliance over understanding, misconduct becomes a rational adaptation to structural pressures rather than an isolated act of immorality.

High-stakes assessments that prioritize recall over application continue to foster punitive and unsustainable learning environments [15, 36]. Under neoliberal governance, where efficiency and performance metrics dominate, universities face mounting pressures—budget constraints, large class sizes, and fragmented curricula—that weaken pedagogical coherence [2]. These conditions amplify uncertainty, pushing both students and educators toward expedient rather than reflective practices. Language assessments vividly illustrate this dynamic: rigid time limits and uniform grading criteria often disadvantage multilingual and international students, who may resort to unauthorized AI tools such as translation or text-generation software to bridge inequities [44]. Taken together, these structural dynamics create the conditions in which AI-mediated shortcuts become particularly attractive, especially when students perceive misalignment between learning goals, assessment expectations, and available support.

Recent empirical studies on AI-mediated learning further demonstrate that integrity cannot be fully understood without accounting for the cognitive, social, and instructional dynamics introduced by human–AI interaction. Research on multi-agent learning systems shows that students increasingly work within distributed networks of human and machine intelligence, where responsibility and authorship are co-constructed rather than individually determined [21, 22]. Emerging evidence on AI-driven feedback ecosystems reveals how real-time automated guidance can reshape cognitive load, metacognitive decisions, and perceptions of task ownership—factors directly linked to integrity choices [33]. Instructional design studies similarly demonstrate that contextualized and adaptive AI supports can inadvertently shift the locus of learning effort, altering when and why students feel compelled to rely on AI rather than their own reasoning

[34]. Parallel research on equity-centered digital learning emphasizes that algorithmic bias, linguistic privilege, and culturally exclusive governance structures can differentially position students within the integrity landscape, reinforcing inequities that shape ethical decision-making [19]. Together, these contemporary studies highlight the urgency of integrating responsible-AI perspectives into academic integrity research to modernize its theoretical grounding and address the realities of AI-enabled higher education. These AI-mediated dynamics amplify pre-existing inequities in access, literacy, and institutional support, linking technological pressures directly to long-standing structural disparities.

Sustainability in education requires more than preventing misconduct—it demands addressing the inequities that make dishonesty seem viable. Unequal access to technology, variable digital literacy, and socio-economic disparities create an uneven ethical terrain where privileged students can leverage AI resources while others face systemic disadvantage [26]. Recent analyses show that digital divides directly shape students' academic outcomes and ethical decision-making, with digitally advantaged students reporting greater confidence and lower perceived risk in using AI tools compared to their peers [5]. These disparities reinforce long-standing patterns of structural inequality and contribute to divergent ethical pressures across student populations. Such asymmetries erode the fairness essential for sustainable learning environments and challenge the moral legitimacy of educational institutions.

Cultural context also shapes the ethical sustainability of learning systems. In collectivist cultures, collaborative practices such as “sharing answers” may reflect solidarity rather than intentional deceit [48]. However, when institutional policies lack clarity or fail to communicate culturally inclusive expectations, these practices blur into academic misconduct. Institutions that neglect to interpret integrity through intercultural lenses inadvertently sustain ethical confusion, fostering climates where misconduct becomes normalized rather than critically examined. Empirical research demonstrates that ambiguous policies and inconsistent enforcement create social environments where cheating becomes normalized within peer groups, as students increasingly perceive dishonest behaviors as common, expected, or low-risk [12, 13]. These culturally situated patterns interact with institutional decision-making, making governance a critical leverage point for restoring trust and coherence.

Governance plays a decisive role in balancing accountability with moral sustainability. Administrative gaps—ranging from inconsistent enforcement of honor codes to excessive surveillance technologies—undermine trust and perpetuate the view of assessment as control rather than collaboration. This tension transforms learning into a transactional process that measures compliance instead of cultivating ethical reasoning. Sustainable governance, by contrast, promotes dialogic accountability, where learners, educators, and policymakers share responsibility for maintaining transparency and trust.

At a deeper level, these pressures mirror the commodification of higher education, where learning is increasingly treated as a market transaction. Empirical studies indicate that students embedded in marketized university systems are more likely to adopt instrumental attitudes toward learning, prioritize credential attainment over engagement, and express higher willingness to engage in dishonest practices when performance stakes are high [3, 46]. In such contexts, ethical sustainability erodes as institutional success becomes decoupled from moral purpose. While some scholars interpret cheating

as an adaptive response to systemic pressures [43], this perspective risks normalizing inequitable and extractive systems that are ethically unsustainable. The expansion of AI technologies, particularly after the COVID-19 pandemic, has further intensified these divides [38]. If left unaddressed, this digital stratification will continue to reproduce ethical and educational inequalities across global HE systems. These systemic pressures illustrate why integrity challenges cannot be solved solely through stricter enforcement; they require structural solutions grounded in ethical stewardship.

Ultimately, reforming academic integrity requires governance structures that align with the United Nations Sustainable Development Goals (SDG 4: Quality Education; SDG 16: Peace, Justice, and Strong Institutions). Ethical sustainability in higher education depends on cultivating institutional ecologies that prioritize inclusivity, moral agency, and shared responsibility over competition and control. By situating cheating within this systemic landscape, integrity is reframed not as a matter of surveillance, but as an outcome of sustainable design, equitable governance, and ethical resilience.

3.1 Positioning the higher education integrity ecosystem model in the literature

Contemporary scholarship has advanced several ecological and systemic accounts of academic integrity that emphasize interactions among policy, pedagogy, and student behaviour (e.g., [1, 7, 18]). Recent systematic reviews and empirical studies further highlight how technology and online assessment have altered these interactions (e.g., [6, 8]). However, many of these frameworks remain oriented toward mapping influences or describing institutional measures rather than rethinking integrity as a sustainability problem that requires reflexive governance and explicit integration of AI ethics [11].

In particular, recent reviews of generative AI's impact on higher education document new forms of ambiguity—where the boundary between legitimate academic support and misconduct is often unclear—and call attention to algorithmic bias and unequal access as drivers of ethical risk [4, 6]. These works argue for policy and pedagogical responses, but they stop short of offering a single integrative model that treats integrity as an emergent property of an institution's sustained alignment among governance, assessment design, culture, and agency.

Another strand of the literature examines how culture building, honor codes, and formative assessment practices can foster integrity as a social norm (Gonsalves, [17]; [32]). Such studies are important demonstrations of the social mechanisms supporting ethical conduct, yet empirical evidence also shows that even well-designed systems can be subverted by deliberate, premeditated cheating—often driven by high stakes pressure, perceived unfairness, or commercial cheating services [20, 24]. This literature challenges purely systems-optimistic views and underscores the need for models that combine preventive design with targeted detection, proportionate sanctioning, and ongoing ethical education.

The Higher Education Integrity Ecosystem Model answers this need by explicitly integrating: (a) AI ethics and technological affordances (to address algorithmic opacity and differential access), (b) reflexive governance mechanisms such as integrity audits and participatory policy design (to ensure continuous adaptation), (c) authentic and process-based assessment (to reduce incentives for shortcutting), and (d) calibrated enforcement and restorative measures (to address deliberate malicious violations). By framing integrity as a sustainability challenge—one that demands both structural design and

responsive governance—the model extends prior ecological accounts and provides an actionable scaffold for institutions facing AI-induced ambiguity.

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Taken together, these strands of research reveal both the potential and the limits of existing systemic approaches. They show that integrity is shaped by assessment design, institutional culture, and governance structures, but they also demonstrate that even well-designed environments remain vulnerable to AI-enabled shortcuts and deliberately orchestrated misconduct. These limitations indicate a need for a more comprehensive framework—one capable of integrating technological disruptions, equity concerns, and the realities of hybrid human–AI cognition. The Higher Education Integrity Ecosystem Model responds to this need by combining preventive design with adaptive governance, supporting both learning and enforcement in a rapidly evolving AI landscape.

4 The higher education integrity ecosystem model

Building on the challenges identified in the preceding sections, this model offers a structured response to the systemic, cultural, and technological pressures reshaping academic integrity in the Age of AI. Rather than treating integrity issues as isolated individual behaviors, the model conceptualizes them as emergent outcomes of interacting institutional layers, each of which is increasingly influenced by generative AI. Given the systemic nature of these pressures, a conceptual framework is required to articulate how the layers interact and where strategic interventions can occur.

This model builds on—but also moves beyond—traditional systemic models of integrity, which typically conceptualize ethical behavior as a balance of policy, pedagogy, and student responsibility [7, 47]). While such frameworks map environmental influences, they rarely consider how technological infrastructures, algorithmic opacity, and digital inequities interact with governance and cultural norms in shaping academic conduct [25, 26]. By contrast, the Higher Education Integrity Ecosystem Model advances the field by explicitly integrating AI ethics, reflexive governance, and sustainability principles. This expanded lens enables a more holistic understanding of integrity as a dynamic governance environment—one that must adapt alongside technological, cultural, and institutional change within higher education. This conceptual gap provides the foundation for a more integrated and sustainability-oriented model, introduced below.

The Higher Education Integrity Ecosystem Model conceptualizes academic integrity as a dynamic, multi-layered system in which governance structures, assessment design, institutional culture, and individual moral agency interact through continuous feedback loops. Rather than treating integrity as a linear outcome of rules or individual morality, the model positions it as an emergent property of a sustainable educational system—one

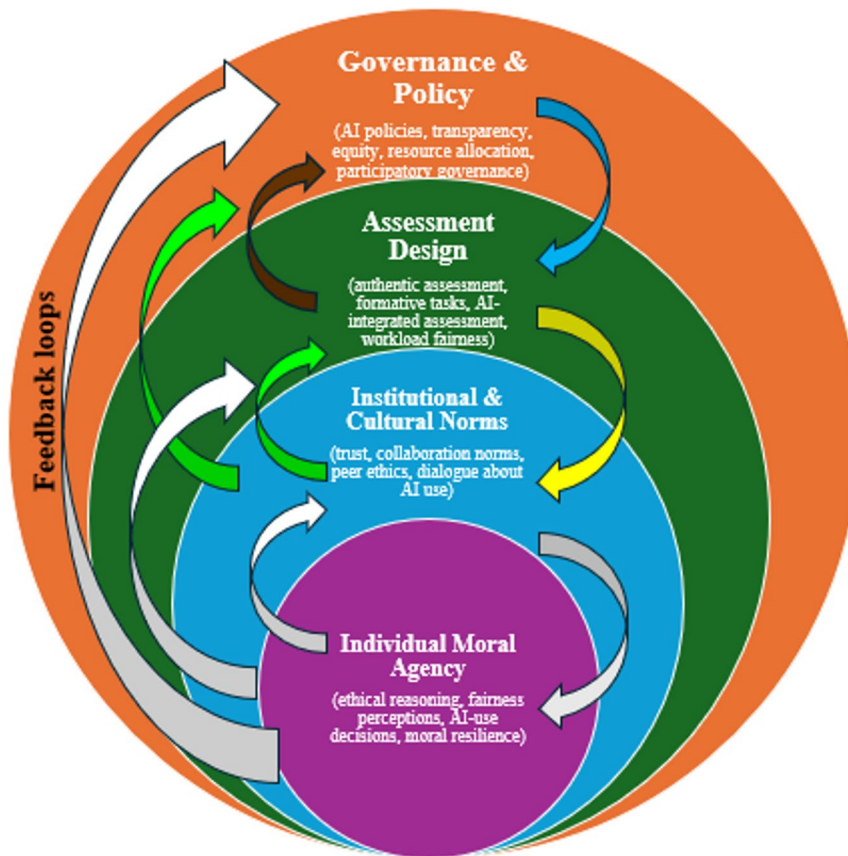


Fig. 1 The higher education integrity ecosystem model

where organizational processes, social norms, and technological infrastructures co-evolve.

To translate this broader conceptual stance into an actionable framework, the model (shown in Figure 1) delineates four interdependent layer: Governance and Policy, Assessment Design, Institutional and Cultural Norms, and Individual Moral Agency. Each layer influences, reinforces, or destabilizes the others. Governance practices—such as transparent AI policies, participatory decision-making, and equitable resource allocation—shape the assessment designs that learners engage with. These pedagogical structures then influence cultural norms around collaboration, fairness, and accountability. Ultimately, these norms shape learners' internalized moral agency, their perceptions of ethical responsibility, and their decision-making in AI-mediated learning environments.

Importantly, the model is explicitly recursive: ethical outcomes feed back into governance adaptation. For example, when integrity violations or equity problems emerge—such as misuse of AI tools or algorithmic disadvantages—these issues serve as inputs for policy revision, assessment redesign, and renewed ethical literacy initiatives. This reflexive loop makes the model responsive rather than static, allowing it to evolve as AI capabilities, educational practices, and learner expectations change. The model therefore provides not only a conceptual lens but also a practical roadmap for constructing systems that support ethical sustainability in higher education.

The model explicitly accounts for the transformative nature of AI by positioning it as an active force that reshapes relationships across all layers of the integrity ecosystem.

At the governance level, AI requires institutions to rethink the meaning of originality, authorship, and verification. At the assessment level, AI compels educators to redesign tasks that cannot easily be outsourced to automated reasoning systems. At the cultural level, AI normalizes hybrid forms of human–machine collaboration that blur ethical boundaries. And at the level of individual agency, AI changes the moral calculus: students must now decide not only whether to cheat, but whether to allow AI to think, write, or evaluate on their behalf. These shifts demonstrate that AI is not an external pressure acting on a stable system—it is a transformative agent that requires the system itself to evolve.

4.1 Interactions and feedback loops across model layers

The Ecosystem Model emphasizes that integrity-related behavior is shaped not by isolated factors but by interactions across system layers. Governance influences Individual Moral Agency by establishing conditions that support or undermine students' ethical development. Clear AI-use guidelines, algorithmic transparency, and institutional fairness practices reduce uncertainty and build trust, which heighten learners' intrinsic motivation to engage honestly. Conversely, punitive surveillance or inconsistent enforcement may erode trust, leading to psychological reactance and greater temptation to cheat.

Assessment Design interacts with Institutional Culture by signaling what the institution values. High-stakes, summative examinations communicate that performance outweighs learning, thereby amplifying competitive norms and normalizing shortcuts. In contrast, authentic, process-based, and formative assessments promote collaboration, reflection, and ethical risk-taking—norms that reinforce sustainable integrity.

Individual Moral Agency is also shaped by Cultural Norms. If peers regularly rely on AI tools beyond institutional policy, students may rationalize similar choices through social proof. Conversely, cultures with transparent dialogue about AI ethics foster peer accountability and moral resilience. These cultural influences then inform Governance, as student feedback, integrity trends, and ethical concerns shape policy revisions and resource decisions.

These multi-directional interactions highlight why integrity cannot be addressed through rules alone: changes to any one layer propagate through the system. The model provides a mechanism for understanding where misalignment occurs and where targeted interventions can restore ethical balance.

The model also incorporates feedback loops that link ethical outcomes to governance adaptation. For example, student feedback collected through integrity audits can inform the redesign of assessment policies, promoting continuous ethical improvement. Likewise, AI ethics training for faculty and students ensures that technological integration remains transparent and equitable. This dynamic adaptability—learning from ethical successes and failures—reflects the sustainability principle of reflexive governance. These recursive interactions become especially significant when examined through the lens of generative AI, which reshapes how each layer functions and interacts.

Framed through this model, academic integrity becomes an emergent property of sustainable systems design. Rather than punishing ethical lapses after they occur, universities can cultivate environments that make ethical engagement the easiest, most natural choice. In essence, the Higher Education Integrity Ecosystem redefines integrity

as a renewable resource—sustained through coherence, trust, and ethical participation across institutional layers (Fig. 1).

The model illustrates four interdependent layers—Governance and Policy, Assessment Design, Institutional and Cultural Norms, and Individual Moral Agency—connected through multi-directional feedback loops. Governance establishes the structural and ethical conditions that shape assessment practices, which in turn influence cultural expectations and norms. These norms shape learners' moral agency and decision-making, while ethical outcomes feed back into governance through evaluation, adaptation, and policy refinement. The model visualizes integrity as an emergent property of the continuous alignment and co-evolution of these system layers.

4.2 Practical illustrations of the ecosystem layers

To move beyond abstraction and demonstrate how the Higher Education Integrity Ecosystem Model operates in practice, it is helpful to illustrate what each layer looks like within real institutional settings. At the governance level, ethical responsibility is enacted through transparent AI-use policies that clearly distinguish acceptable support from misconduct. For instance, a university may explicitly permit AI for grammar correction but prohibit full-text generation, pairing this with institution-wide workshops that explain the rationale behind these policies. Ethical reciprocity becomes evident here: the institution's commitment to clarity and fairness encourages students to act honestly because they understand the rules and feel included in their development.

Within the assessment design layer, integrity-supportive practices take the form of authentic, process-oriented tasks that make misconduct less attractive and learning more meaningful. A concrete example is the use of multi-stage assignments—such as proposal → draft → peer feedback → final submission—where each stage invites reflection and reduces the incentive to misuse AI for quick completion. Ethical reciprocity is demonstrated when instructors design assessments that prioritize learning, and students reciprocate by engaging with these stages rather than outsourcing the work to external tools.

Institutional and cultural norms become visible through the everyday practices that shape how students perceive ethical behavior. For example, student-led discussion forums or “AI Integrity Circles” can facilitate open dialogue about responsible AI use across different disciplines. When students collectively agree to limit certain forms of AI assistance—such as using AI only for brainstorming rather than full drafting—they actively reinforce a culture of transparency and shared responsibility. This cultural dimension also reflects ethical reciprocity: the institution normalizes open conversations about integrity, and students contribute by shaping and sustaining those norms.

At the innermost layer, individual moral agency appears in small but meaningful decisions students make during their learning process. A student might choose, for instance, to limit AI assistance because they recognize that over-reliance weakens their ability to learn the skill being assessed. Reflective statements such as “Which parts of this assignment did I use AI for, and why?” foster internal ethical reasoning and allow students to articulate their decision-making processes. This demonstrates ethical reciprocity by linking institutional efforts to build ethical capacity with students' personal commitment to act responsibly.

These examples illustrate how the model operates across policy, pedagogy, culture, and agency. They show how ethical reciprocity is not an abstract idea but a lived practice in which institutions design trustworthy systems and learners respond with ethical engagement. The result is an integrated environment where integrity emerges from alignment between structural supports and individual moral decisions.

Importantly, the Age of AI forces institutions to move beyond incremental adjustments toward systemic redesign. AI's capacity to generate coherent academic output, simulate disciplinary writing styles, and provide instant feedback means that traditional integrity safeguards—such as plagiarism detection, honor codes, or proctoring—are no longer sufficient. Ethical sustainability now requires aligning policies, assessment practices, cultural norms, and moral agency with a technological environment in which machine assistance is ubiquitous and increasingly indistinguishable from human production. The ecosystem model provides a structure for this systemic adaptation, ensuring that integrity remains viable in an era where the boundaries between human and AI contributions are fluid and continually evolving.

4.3 How generative AI challenges each layer of the integrity ecosystem

Generative AI presents not only procedural challenges but conceptual ones: it reshapes the meaning of authorship, originality, and ethical responsibility across all layers of the ecosystem. Rather than merely “blurring lines,” AI alters the internal logic of governance, assessment, culture, and agency. This subsection offers a more granular analysis of how generative AI pressures each layer of the model and requires reconsideration of long-standing educational assumptions.

1. **Governance & Policy:** AI destabilizes the foundations of authorship and verification
Generative AI tools challenge governance by undermining the basic assumptions that integrity policies traditionally rely on—namely, that human authorship is observable, verifiable, and distinguishable from external assistance. When a student can produce a polished essay, dataset summary, or problem-solving explanation using minimal prompts, institutions must redefine what qualifies as “student work.” Traditional enforcement mechanisms such as plagiarism detection become insufficient because AI-generated content is often original by design [16]. As a result, governance must shift from content-based verification (“Did the student produce this text?”) to process-based accountability (“How did the student produce this text, and with what tools?”). This transformation forces institutions to rethink authorship criteria, develop transparent AI-use classifications, and redesign integrity policies around process, disclosure, and intent rather than solely around textual originality.
2. **Assessment Design:** AI disrupts the meaning of authenticity and task validity
AI fundamentally challenges assessment design because many traditional “authentic” tasks—essays, literature reviews, summaries, reflective writing, and even some forms of problem-solving—can now be completed by AI at or above human-level proficiency [41]. This means authenticity can no longer be defined simply as tasks that resemble real-world practice; instead, authenticity must be reconceptualized as tasks that require human judgment, embodied experience, personal insight, contextual reasoning, ethical deliberation, or situated interaction that AI cannot replicate. For example, reflective narratives anchored in classroom discussions, oral defenses of written work, or tasks requiring iterative development with instructor

feedback become more meaningful indicators of student learning. Generative AI thus forces assessment design to evolve from evaluating final products to evaluating learning processes, interactions, and human decision-making—shifting the locus of authenticity from output to engagement.

3. Institutional & Cultural Norms: AI reshapes collective expectations about learning and collaboration

Generative AI normalizes forms of hybrid cognitive labor in which human and machine contributions are intertwined [14]. This disrupts cultural norms that once drew clear boundaries between “support” and “outsourcing.” For example, peer groups may begin to see AI-assisted brainstorming, translation, or rewriting as benign academic habits, even when these practices conflict with institutional policies. As AI becomes embedded in everyday life—writing emails, generating ideas, summarizing texts—it becomes culturally difficult to justify why academic tasks should be exempt from such assistance [45]. This tension forces institutions to cultivate new norms of transparency, disclosure, and responsible use rather than relying on pre-AI cultural expectations. AI therefore acts as a cultural accelerator, rapidly shifting norms of collaboration, intellectual ownership, and the value of human effort.

4. Individual Moral Agency: AI complicates ethical decision-making by redistributing cognitive labor

At the innermost layer, generative AI directly alters what it means for students to exercise moral agency. In pre-AI contexts, integrity decisions centered primarily on whether to cheat; now, learners must decide how much intellectual work they will delegate to machines [42]. Moral agency in the Age of AI involves questions such as: “Should I let AI propose my thesis argument?”, “Is relying on AI for interpretation preventing me from developing my own understanding?”, or “Does using AI for feedback compromise my authentic learning?” These dilemmas are not simply acts of dishonesty—they are negotiations of identity, responsibility, and authorship. Agency becomes a matter of ethical self-regulation in the face of highly capable tools [30]. Thus, AI does not only influence students’ choices, it reshapes the very meaning of choice by changing the distribution of cognitive effort. Integrity becomes less about resisting cheating and more about deciding when, why, and to what extent to involve AI in one’s intellectual development.

4.4 Synthesis: AI as a conceptual disruptor, not only a procedural challenge

Across all layers of the ecosystem, generative AI does not merely complicate existing practices—it transforms the underlying assumptions that make those practices meaningful. It forces governance to redefine authorship, compels assessment design to develop new forms of authenticity, pushes cultural norms toward hybrid human-AI collaboration, and expands moral agency into a new form of ethical self-regulation. This deeper, systemic analysis explains why AI is not just a pressure acting on the system but a catalyst that requires the system itself to evolve.

4.5 Limitations and boundaries of the ecosystem model

While the Higher Education Integrity Ecosystem Model offers a holistic framework for understanding how governance, assessment design, culture, and moral agency interact, it is important to acknowledge its limitations and the conditions under which it may fail

to operate effectively. Because the model depends on alignment across multiple institutional layers, it is most successful in contexts where governance structures are stable, resources are sufficient, and faculty and students have the time and support to engage with integrity-driven practices. In environments characterized by fragmented leadership, inconsistent policy communication, or rapid organizational turnover, the feedback loops essential to the model's adaptability may weaken or break entirely.

A significant limitation of the model lies in its reliance on institutional capacity. Implementing transparent AI policies, conducting integrity audits, redesigning assessments, and supporting cultural development require resources, ongoing training, and technical infrastructure. Under-resourced institutions may struggle to maintain these components simultaneously. For example, universities with large student–faculty ratios or limited digital access may not be able to support authentic assessment redesign or sustained ethics education. In such contexts, the ecosystem risks becoming aspirational rather than operational, as systemic constraints undermine the conditions necessary for ethical sustainability.

Political and cultural constraints also shape the model's boundaries. In institutions where academic integrity is framed narrowly as rule enforcement, or where decision-making is centralized and non-participatory, the collaborative governance needed for ethical reciprocity may be difficult to achieve. Power asymmetries, fear of punitive consequences, or mistrust between students and administration can weaken the cultural layer of the ecosystem, causing the model to underperform despite strong policies on paper. Additionally, in sociopolitical environments with restricted academic freedom or limited space for open dialogue, efforts to cultivate integrity-driven norms may be constrained or actively resisted.

The model also faces limitations in addressing deliberate, highly organized, or commercialized forms of misconduct. Although it supports the prevention of system-induced or opportunistic cheating, it cannot fully eliminate calculated breaches involving contract cheating services, identity fraud, or sophisticated AI misuse. These threats may persist even when governance, assessment design, and cultural norms are well aligned. In such cases, the model requires complementary safeguards such as identity verification technologies, targeted detection mechanisms, and sector-level policy coordination.

Finally, the model assumes that institutions are willing to reflect, adapt, and revise policies based on feedback—an assumption that may not hold universally. Without reflexive governance, feedback loops collapse, and integrity challenges may accumulate despite positive efforts at the classroom or cultural level. Thus, while the ecosystem model offers a comprehensive and sustainable approach, its success ultimately depends on political will, institutional transparency, and the availability of resources to support continuous ethical development.

5 Toward sustainable integrity: policy and practice implications

Building a sustainable culture of integrity requires reimagining governance as stewardship rather than surveillance. Institutions must transition from reactive enforcement toward proactive system design that embeds equity, transparency, and dialogue at every level of assessment. The HE Integrity Ecosystem offers a roadmap for such transformation by connecting ethical theory with practical governance strategies that align with the Sustainable Development Goals (SDG 4 and SDG 16).

At the policy level, universities should institutionalize annual integrity audits—evaluations of assessment fairness, AI use, and equity alignment. These audits can be informed by anonymous student surveys, classroom-level feedback, and peer review mechanisms, ensuring that policies evolve through participatory dialogue. Embedding integrity metrics into sustainability reports would also reinforce accountability and signal that ethical governance is central to institutional performance.

To operationalize integrity in the Age of AI, policy recommendations must move beyond general formulations of “ethical AI literacy” and toward concrete, skills-based strategies. An effective AI-integrity curriculum would include modules on prompt provenance (how prompts shape and bias outputs), authorship transparency (when and how to disclose AI involvement), critical evaluation of AI outputs (including fact-checking, hallucination detection, and bias identification), and co-creation with AI (how to integrate AI assistance while preserving human reasoning and disciplinary knowledge). Such a curriculum should also teach students how to document the stages of AI interaction during assignment development, creating an auditable trail of decision-making that reinforces accountability.

At the assessment level, AI-integrity strategies require institutions to articulate discipline-specific norms for acceptable AI use. For example, in essay writing, policies should clarify which types of assistance are permissible—such as idea brainstorming, language polishing, or structural suggestions—and which are prohibited, such as generating full paragraphs, arguments, or citations. The ecosystem model supports this process by highlighting how governance (policy clarity), assessment design (task purpose), cultural norms (shared expectations), and individual agency (self-regulation) must all align for the policy to function. An institution implementing such a policy might require students to submit an “AI Use Statement” that describes what tools were used, for what purpose, and under which constraints, paired with occasional oral defenses or reflective notes that verify authorship and understanding.

More broadly, the model guides institutions in differentiating between low-risk and high-risk AI use cases. Low-risk uses—such as grammar checking, accessibility tools, or summarization of student-generated content—can be integrated into policy with minimal restrictions. High-risk uses, such as generating solutions for coding assignments or writing entire analytical essays, require targeted guidance, transparency practices, and redesigned assessment formats. By mapping these risk levels onto the four ecosystem layers, institutions can develop coherence: governance sets clear boundaries, assessment design renders misconduct less attractive, cultural norms promote disclosure, and students develop the moral agency to navigate AI’s capabilities responsibly.

At the pedagogical level, faculty should integrate peer-review training and formative assessment practices that emphasize collaboration over competition. These approaches distribute ethical responsibility across learning communities, enabling students to practice evaluative judgment and moral reasoning within authentic academic contexts. Such participatory learning strategies not only reduce opportunities for misconduct but also promote the psychological resilience essential for ethical sustainability.

At the technological level, institutions must embrace ethical AI literacy. This entails semester-long professional development for educators and workshops for students that address responsible AI use, transparency in algorithmic feedback, and bias awareness.

By embedding AI ethics into everyday academic practice, universities can mitigate digital inequities and strengthen trust in emerging technologies.

Ultimately, sustainable academic integrity depends on recognizing ethics as a collective, renewable resource. Governance must lead by embedding assessment literacy and integrity education into institutional policy, ensuring that ethical capacity grows with technological and cultural change. The cultivation of integrity should not rest on surveillance or deterrence, but on the co-creation of just and trustworthy systems.

Integrity in higher education is, therefore, an indicator of sustainability itself. It reflects how institutions balance fairness, innovation, and accountability while nurturing future citizens capable of ethical decision-making in complex global environments. By adopting the Higher Education Integrity Ecosystem Model, universities can move beyond the punitive logic of misconduct toward a sustainability-centered paradigm—one in which ethical behavior, like environmental stewardship, is continuously renewed through shared responsibility, transparency, and care.

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